

22 Appendices

22.1 Lycra suit sizes overview

The typical height and weight of a subject that will fit into a given suit size is related. While this is generally accurate, it is an indication, therefore test the suit size, to ensure that it is tight enough to keep MTx's in place during measurement while remaining relatively comfortable.

Size	Body height		Weight	
S	155 - 165 cm	5'1" - 5'4" ft.	50 - 65 kg	110 - 143 lbs.
M	165 - 170 cm	5'4" - 5'6" ft.	65 - 75 kg	143 - 165 lbs.
L	170 - 180 cm	5'6" - 5'9" ft.	75 - 85 kg	165 - 187 lbs.
XL	180 - 195 cm	5'9" - 6'4" ft.	85 - 95 kg	187 - 210 lbs.
XXL	195 - 205 cm	6'4" - 6'8" ft.	95 - 105 kg	210 - 231 lbs.
XXXXL	205 - 215 cm	6'8" - 7'1" ft.	105 - 115 kg	231 - 254 lbs.

22.2 MVN kinematics and output

22.2.1.0 Quaternion orientation representation

A unit quaternion vector can be interpreted to represent a rotation about a unit vector $\mathbf{n}$ through an angle α .

$${}^{GB}q = (\cos(\frac{\alpha}{2}), \mathbf{n} \sin(\frac{\alpha}{2}))$$

A unit quaternion itself has unit magnitude, and can be written in the following vector format:

$$q = (q_0, q_1, q_2, q_3)$$

$$\|q\| = 1$$

Quaternions are an efficient, non-singular description of 3D orientation and a quaternion is unique up to sign:

$$q = -q$$

An alternative representation of a quaternion is as a vector with a complex part, the real component is the first element q_0 .

The inverse ${}^{BG}q$ is defined by the complex conjugate * of ${}^{GB}q$:

$${}^{GB}q^* = (q_0, -q_1, -q_2, -q_3) = {}^{BG}q$$

As defined here, ${}^{GB}q$ rotates a vector ${}^B\mathbf{x}$ in the body co-ordinate system (B) to the global reference co-ordinate system (G).

$${}^G\mathbf{x} = {}^{GB}q \otimes {}^B\mathbf{x} \otimes {}^{GB}q^* = {}^{GB}q \otimes {}^B\mathbf{x} \otimes {}^{BG}q$$

Where $\otimes$ represents a quaternion multiplication:

$$q_1 \otimes q_2 = (q_{10} \cdot q_{20} - v_1 \cdot v_2, q_{10} \cdot v_2 + q_{20} \cdot v_1 + v_1 \times v_2)$$

with:

$$v_1 = (q_{11}, q_{12}, q_{13})$$

$$v_2 = (q_{21}, q_{22}, q_{23})$$

and both $\cdot$ and $\times$ represent the standard dot and cross product, respectively. Be aware that the order of multiplication is important. Quaternion multiplication is not commutative, meaning:
 $q_1 \otimes q_2 \neq q_2 \otimes q_1$

22.2.2.0 Conversions

Quaternion to rotation matrix:

$$\mathbf{R} = \begin{bmatrix} q^2 + q_1^2 - q_2^2 - q_3^2 & 2q_1q_2 - 2q_3q_0 & 2q_1q_3 + 2q_2q_0 \\ 2q_1q_2 + 2q_3q_0 & q^2 - q_1^2 + q_2^2 - q_3^2 & 2q_2q_3 - 2q_1q_0 \\ 2q_1q_3 - 2q_2q_0 & 2q_2q_3 + 2q_1q_0 & q^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2q_1q_2 - 2q_3q_0 & 2q_1q_3 + 2q_2q_0 \\ 2q_1q_2 + 2q_3q_0 & 1 - 2q_1^2 - 2q_3^2 & 2q_2q_3 - 2q_1q_0 \\ 2q_1q_3 - 2q_2q_0 & 2q_2q_3 + 2q_1q_0 & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix}$$

For further information on quaternion operations and rotations, refer to the book: *J.B. Kuipers. Quaternions and Rotation Sequences. Princeton University.*

a. Body planes

Sagittal plane

Frontal / coronal plane

Transverse plane

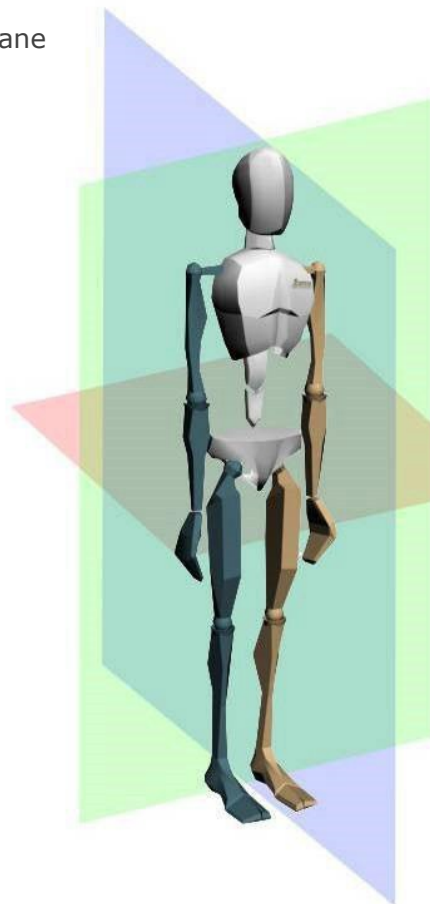


Figure 90: Body planes

b. Coordinate systems

The MVN Fusion Engine calculates the position and orientation, and other kinematic data of each body segment, B , with respect to an earth-fixed reference co-ordinate system, G . By default, the earth-fixed reference co-ordinate system used is defined as a right-handed Cartesian co-ordinate system with:

Global reference frame:

- i. X positive when pointing to the local magnetic North (red).
- ii. Y according to right-handed co-ordinates (West) (green).
- iii. Z positive when pointing up (blue).

The body frames are aligned with the Global reference frame when the subject is standing in T-pose. For the calculation of joint angles, an extra frame of reference (based on anatomical pose) is used to define the body frames. This frame of reference defines the body frames by:

1. Origin: center of rotation (proximal)
2. X forward.
3. Y up, from joint to joint.
4. Z pointing right.

This different coordinate system is **only** used as an intermediate frame to calculate joint angles. Joint origins are determined by the anatomical frame and are defined in the center of the functional axes with the directions of the X, Y and Z being related to functional movements. For example, flexion/extension of the knee is described by the rotation about the BZ -axis of the lower leg with respect to the upper leg; abduction/adduction is the rotation about the BX -axis; and endo/exo rotation is about the BY -axis. Note that in some studies, different anatomical and reference frames definitions are used; to compare results proper co-ordinate frame transformations must be performed.

After the calibration procedure, the right heel of the character is set at the origin (0,0,0). This is the begin point of the red arrow in the 3D viewport (pointing in the positive X-direction).

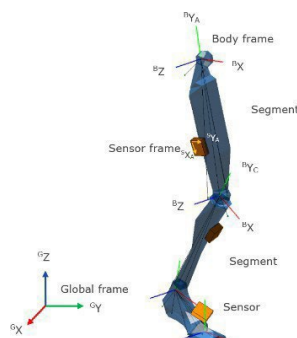


Figure 91: Global and local coordinate frames in Xsens Analyze/Animate for the calculation of joint angles

c. Anatomical model

22.2.1.0 Definition of segment axes

To describe motion capture data in clinically meaningful data, e.g. joint rotations, it is necessary to define a coordinate reference frame attached to the body segment. The ISB proposal for standardization for joint coordinate systems is currently based on positions (optical markers) of bony landmarks. For most segments on the body, the origin of the reference frame and direction of axes are described in Wu and Cavanagh (1995) ^[i] Wu et al. (2002) ^[ii] and Wu et al. (2005) ^[iii]. The MVN Fusion Engine uses the calibration poses to determine the direction of the axes of each segment. Figure 92a shows the origins of the segment axes in the neutral anatomical pose. The origin of the segment is always in the proximal joint center which is the functional rotation point. The origin is chosen such that by connecting all joint centers a character (skeleton) is formed, see Figure 92b. Their positions for the calibration poses in the global reference frame can be found in the <segments>section and <frame type="tpose/tpose-isb"> when exporting as MVNX file.

22.2.4.0 Bony/anatomical landmarks

The positions of bony/anatomical landmarks with respect to the joint origin of the related segment are presented in the MVNX file, see the <points>tag in the <segment>section. Segments are assumed to be rigid bodies.

Positions of anatomical landmarks in the global frame (G) can be found by rotating the vector of the landmark in the body frame (B) to the global frame and adding the global position of the origin.

$${}^G \mathbf{p}_{\text{landmark}} = {}^G \mathbf{p}_{\text{origin}} + {}^{GB} \mathbf{q} \otimes {}^B \quad {}^{GB} *$$

Or in rotation matrix formulation:

$${}^G \mathbf{p}_{\text{landmark}} = {}^G \mathbf{p}_{\text{origin}} \quad {}^{GB} \quad {}^B$$

The rotation from body to global (${}^{GB}\mathbf{q}$) and global position of the origin (${}^G\mathbf{pos}$) are given in the MVNX file, see next section.

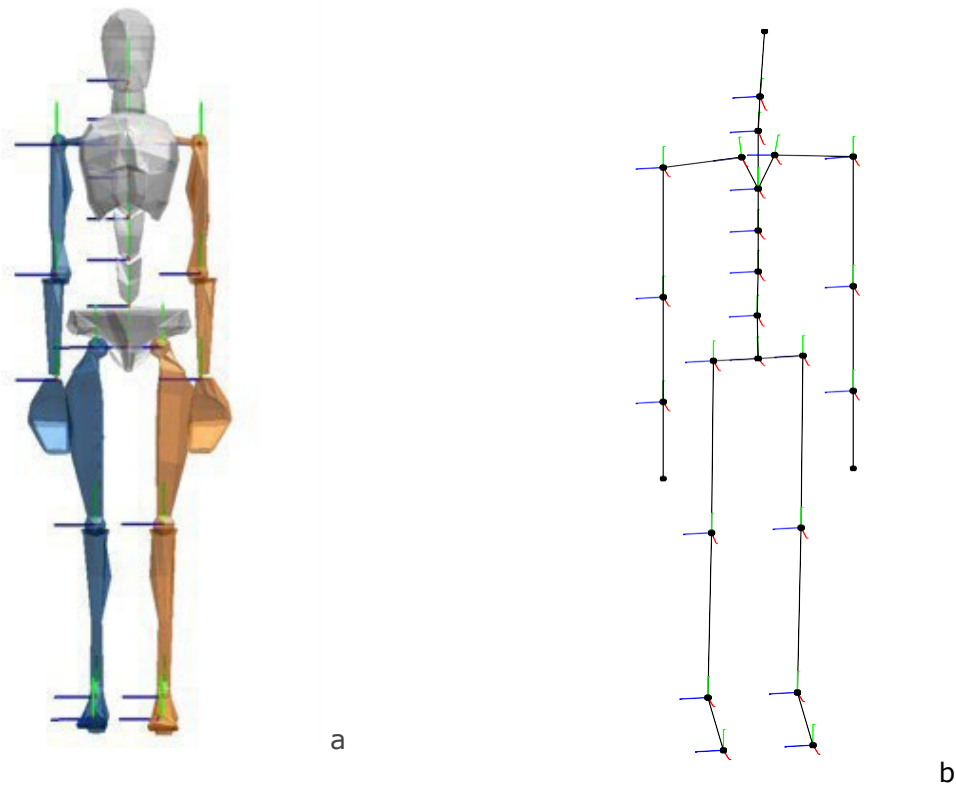
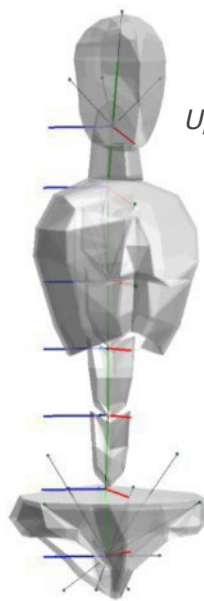


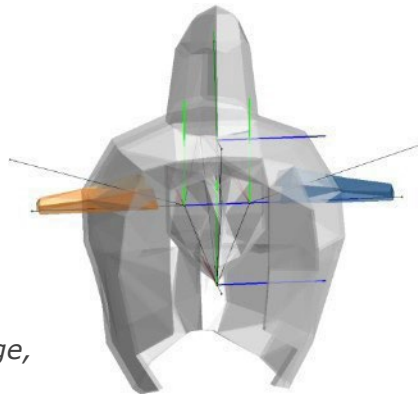
Figure 92: a) Segment coordinate system at each joint origin, as is used in MVN to determine joint angles. b) Joint centers connected, forming a skeleton Legend: **x: red, y: green, z: blue**

Note that positions of anatomical landmarks are not measured directly as they are with optical measurement systems. They are computed using the measured accelerations, angular velocities and rotations in combination with the biomechanical model. Relevant points are used for the detection of contacts with the external world and as an indication for scaling of segment lengths.

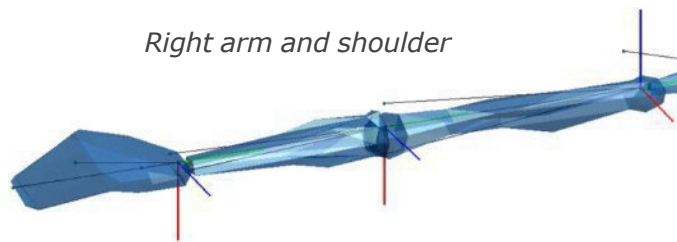
The MVN character mesh (only for visualization) is an abstract representation of the human skeleton; therefore, not all anatomical landmarks are exactly on the mesh. The ribcage is connected to T8 and will move accordingly.



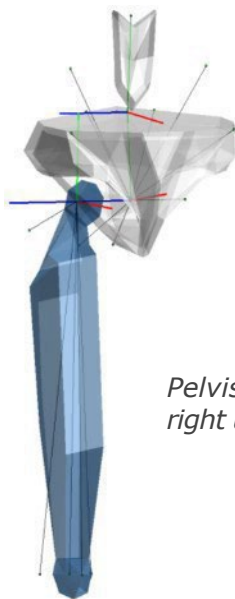
Upper body



*T8, ribcage,
neck and
shoulders*



Right arm and shoulder



*Pelvis, L5 and
right upper leg*



*Left leg and
foot*

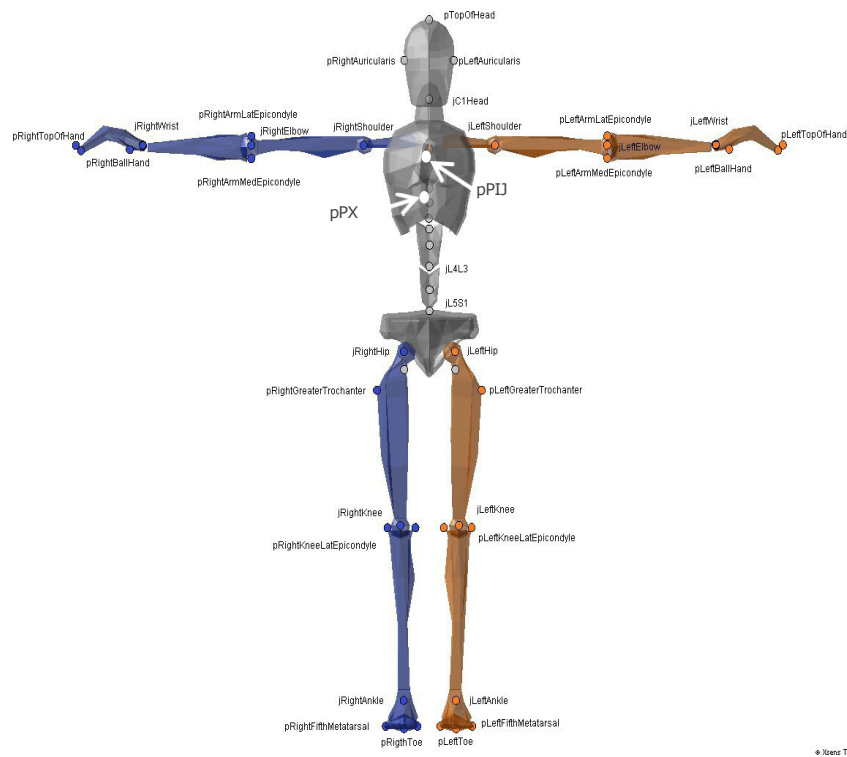
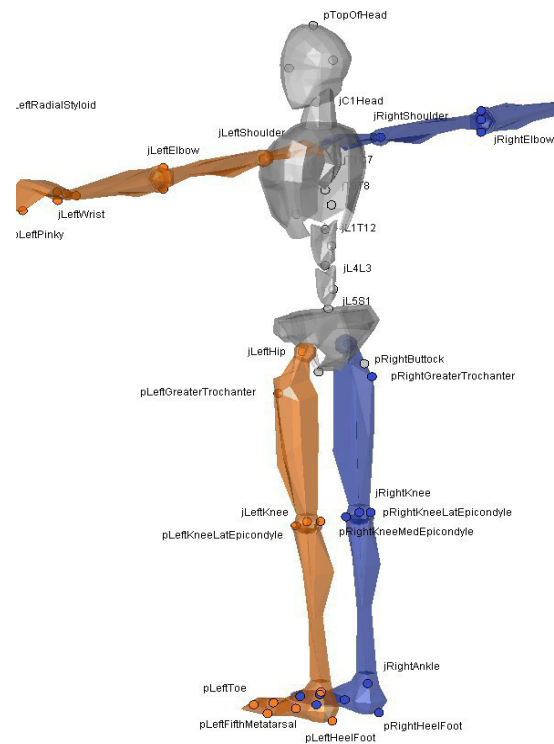


Figure 93: Selection of anatomical landmarks. Above, anterior view. Below, posterior view.

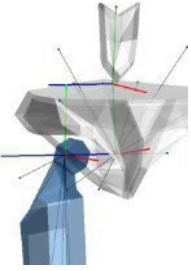


d. Segment axes definitions and Origin Definitions

Note: To ease comparison between the MVN segment co-ordinate systems with the International Society of Biomechanics recommendation, currently based on optical

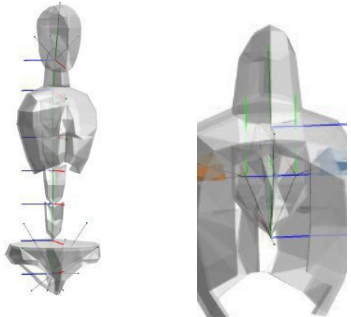
position tracking systems, the ISB recommendation is presented next to the MVN definition.

The MVN segment axes definition and origins have been chosen as closely as possible to the ISB recommendation where whenever possible.

MVN		ISB [ij]
	O	Midpoint between right and left hip center of rotation Coincident with hip center of rotation
	X	Perpendicular to Y and Z Pointing forward Line parallel to a line lying in the plane defined by two ASISs and midpoint of two PSIS, orthogonal to Z-axis. Pointing anteriorly.
	Y	hipOrigin to jL5S1 Pointing up Line perp. to X and Z. Pointing cranially. (Della Croce et al.[iv])
	Z	to jLeftHip to jRightHip Pointing right Line parallel to a line connecting right and left ASISs. Pointing right.

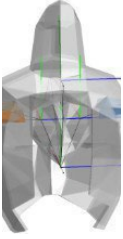
NB: Della Croce et al. [iv] define O as the midpoint between RASIS and LASIS

22.4.1.0 Spinal segments: L5, L3, T12, T8: Segments 2-5

MVN		ISB [ij]
	O	jL5S1, jL4L3, jL1T12, jT9T8
	X	Pointing forward Line per. Y and Z
	Y	Joint to Joint Pointing up Line passing thru centers of vertebra's upper and lower endplates
	Z	Perp. to X and Y Pointing right Line parallel to a line joining similar landmarks on the bases and right and left pedicles

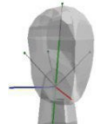
Note: Spine segments not measured directly in MVN, they are interpolated between the MT's of the Pelvis, Sternum and Head using a model of the spine. Joint origins are the rotation points in the vertebrae.

22.4.2.0 Neck: Segment 6

MVN		ISB
	O	jT1C7
	X	Perp. to Y and Z Pointing forward
	Y	jT1C7 to jC1Head
	Z	Pointing right
		Not described in ISB

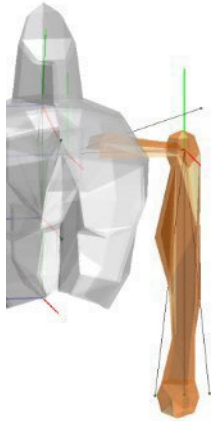
Note: The neck segment is not measured directly in MVN; it is calculated using a model of the neck.

22.4.3.0 Head: Segment 7

MVN		ISB
	O	jC1Head
	X	Perp. to Y and Z
	Y	jC1Head to TopOfHead
	Z	Left Ear to Right Ear
		Not described in ISB

Note: Position of Ears are not measured but assumed to be on the same height.

22.4.4.0 Shoulder: Segment 8 Right and Segment 12 Left

MVN		ISB
	O	jRightC7Shoulder jLeftC7Shoulder
	X	Pointing forward
	Y	Perp. to X and Z
	Z	Right: jRightC7Shoulder to jRightShoulder to Left: jLeftC7Shoulder to jLeftC7
		See Wu et al. [iii] and Šenk et al. v for range of definitions.

In ISB the shoulder complex is described with the Thorax, Clavicle, Scapula and Humerus. MVN does not define the thorax segment. The thorax region is split into spine segments. The mesh of the thorax will move with segment T8 which is measured with the MT of the sternum. MVN does not measure the clavicle and is undefined in MVN.

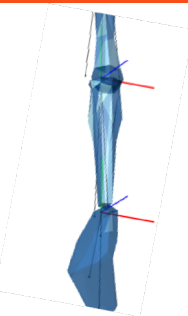
22.4.5.0 Upper Arm (Humerus): Segment 9 Right and Segment 13 Left

There are two segment coordinate systems described in ISB. The system described below is option 1.

MVN		ISB [iii]
	O	jRightUpperArm GH jLeftUpperArm GH
	X	Sagittal plane Pointing forward
	Y	Right: jRightElbow to jRightShoulder Left: jLeftElbow to jLeftShoulder
	Z	Perp. to X and Y Pointing to the right

22.4.6.0 Forearm (Radius/Ulna): Segment 10 Right and Segment 14 Left

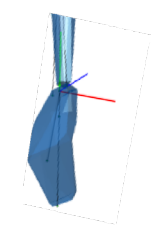
There are two segment coordinate system described in ISB. Option 1 is described here.

MVN		ISB [iii]
	O	jRightElbow jLeftElbow
	X	Pointing forward
	Y	Right: jRightWrist to jRightElbow Left: jLeftWrist to jLeftElbow
	Z	Perp. to X and Y Pointing right

Note: The joint origin is the midpoint between the epicondyles in MVN.

In the anatomical pose, the hand and forearm are orientated forward

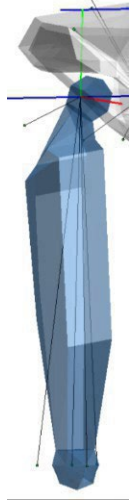
22.4.7.0 Hand: Segment 11 Right and Segment 15 Left

MVN		ISB [iii]
	O	jRightWrist jLeftWrist
	X	Pointing forward
	Y	Top of Hand to jRightWrist/jLeftWrist Pointing Vertical
	Z	Perp. to X and Y Pointing right

Note: In MVN, the joint center is the midpoint between the styloids. No metacarpals are measured in MVN.

This description is for the anatomical pose. When the subject stands in the N-pose, X will no longer point anteriorly, but medially.

22.4.8.0 Upper Leg (Femur): Segment 16 Right and Segment 20 Left

MVN		ISB [ii]
	O	jRightHip/jLeftHip (Only defined for use of calculating the hip joint angle): Coincident with right (or left) hip center of rotation, coincident with pelvic center O in neutral configuration
	X	Perpendicular to Y and Z Pointing forward in sagittal plane Line perp. to Y- and x- Pointing anteriorly. (Della Croce et al. [iv])
	Y	Right: jRightKnee to jRightHip Left: jLeftKnee to jLeftHip Line joining midpoint between medial and lateral Fes and the origin. Pointing cranially
	Z	Right: Medial to Lateral Left: Lateral to Medial Pointing right Line perp. to Y-axis lying in plane defined by origin and two Fes. Pointing Right.

Note: In ISB and MVN, the origin is in center of hip rotation, which is the head of the femur. Della Croce [iv]: midpoint LE and ME. In the MVN model, the lateral and medial epicondyles are on the same height.

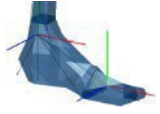
22.4.9.0 Lower Leg (Tibia/Fibula): Segment 17 Right and Segment 21 Left

MVN		ISB [iii]
	O	jRightKnee jLeftKnee IM
	X	Perp. Y and Z Pointing forwards Line perp. to torsional plane of tibia/fibula. Pointing anteriorly
	Y	Right: jRightAnkle to jRightKnee Left: jLeftAnkle to jLeftKnee Line perp. to X and Z
	Z	Right: Medial to Lateral Left: Lateral to Medial Pointing right Line between MM and LM. Pointing to the right

Note: In MVN, the joint center is approx. between LC and MC which corresponds with the center of rotation of the lower leg. Same coordinate system as Della Croce [iv], except that the origin is not in IM. In ISB, the Y axis is not along the

Tibia. In the MVN model the lateral and medial epicondyles and lateral and medial malleoli are the same height.

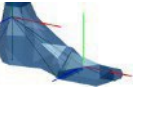
22.4.10.0 Foot (Calcaneus): Segment 18 Right and Segment 22 Left

MVN			ISB [iii]
	O	jRightAnkle jLeftAnkle IM	IM
	X	Sagittal plane forward	Line perp. to frontal plane of tib/fib in neutral configuration. Pointing anteriorly.
	Y	Vertical (aligned with gravity, pointing up)	Line coincident with long axis of tibia/fibula, in neutral configuration. Pointing cranially
	Z	Perp. to X and Y Pointing right	Line perp. to x- and Y-

Note: In the MVN model, lateral and medial malleoli are on the same height.

22.4.11.0 Toe: Segment 19 Right and Segment 23 Left

The toes are not defined in ISB.

MVN			ISB
	O	jRightToe jLeftToe	The toes are not defined in ISB.
	X	Sagittal plane forward	
	Y	Vertical (aligned with gravity)	
	Z	Perp. to X and Y	

Note: In MVN toes are not measured directly but are based on the foot kinematics and contact detection. The origin is between the metatarsals of the foot. Toes can only flex and extend, other kinematics are due to interpolation.

e. Joint angles

Typically, a joint rotation is defined as the orientation of a distal segment ${}^{GB}q$ with respect to a proximal segment ${}^{GB_A}q$:

$${}^{B_A B_B}q = {}^{GB_A}q^* \otimes {}^{GB_B}q$$

where $\otimes$ denotes a quaternion multiplication and * the complex conjugate of the quaternion (see Section 23.2.1). There are a few commonly used parameterizations of the joint rotation ${}^{B_A B_B}q$ that describe joint angles:

- the Cardan/Euler representation [Cole 1993, Woltring 1994]
- joint coordinate system [Grood & Suntay 1983]^{vi}
- helical angle [Kinzel 1972]

Note that all representations of the joint angles are based on the same quaternion or rotation matrix. The differences lie only in how the angles are extracted (represented) from this rotation. The International Society of Biomechanics (ISB) has proposed

standards for rotation sequences for the lower [Wu et al, 2002ii] and upper body [Wu et al 2005iii].

22.5.1.0 Euler Extractions for the joint angles

In order to produce validated joint angle output, Xsens MVN works closely to the ISB and Grood and Suntay recommendations [i,ii,iii,vi]. Almost all angles follow the ISB Euler angle extractions of Z (flexion/extension), X (abduction/adduction) Y (internal/external rotation). The definitions of the origins of the segments are somewhat different, since MVN uses MT's placed on the segment, rather than markers placed on bony landmarks. The origins and coordination axes are defined above.

22.5.2.0 Shoulder angle definitions

For joint angle calculations, the only difference for Euler extractions between MVN and ISB is for the shoulder joint. MVN provides three different shoulder angle calculations, as can be seen in the data tree, after a session has been recorded.

The shoulder joint angles provided are:

- Angles between C7 and the shoulder segment
- Euler rotation sequence ZXY
- Euler rotation sequence XZY

22.5.3.0 Joint angle outputs

Below is an overview of all joints available in MVN and a short description.

Joint Angle		Description (& Euler Sequence)
1	L5S1	Joint between the lumbar spine segment 5 and sacral spine 1 (ZXY)
2	L4L3	Joint between the lumbar spine segment 4 and lumbar spine segment 3 (ZXY)
3	L1T12	Joint between the lumbar spine segment 1 and thoracic spine segment 12 (ZXY)
4	C1Head	Joint between the cervical spine segment 1 and the head segment (ZXY)
Left and Right		
5	C7Shoulder	Joint between cervical spine 7 and the MVN shoulder segment
6	Shoulder ZXY	Shoulder joint angle between the MVN shoulder segment and the upper arm; calculated using the Euler sequence ZXY
7	Shoulder XZY	Shoulder joint angle between the MVN shoulder segment and the upper arm; calculated using the Euler sequence XZY
8	Elbow	Joint between the upper arm and the forearm. (ZXY)
9	Wrist	Joint between the forearm and the hand. (ZXY)
10	Hip	Joint between the pelvis and upper leg. (ZXY)
11	Knee	Joint between the upper leg and lower leg. (ZXY)
12	Ankle	Joint between the lower leg and foot. (ZXY)
13	BallFoot	Joint between the foot and the calculated toe. (ZXY)

f. Xsens motion capture engine

Traditional motion capture systems based on inertial and magnetic sensing typically estimate the position and orientation of each segment by applying the result of a sensor-

to-segment calibration procedure to an estimate of sensor orientation and applying a (scaled) biomechanical model of the human body. Each of these components will introduce errors, but the most dominant error sources are generally the magnetic distortions which are present anywhere (e.g. common materials in buildings, furniture, and surrounding electronic equipment). The typical sensor fusion of inertial and magnetic sensor signals is done by combining the tracking of short-term changes in orientation by the gyroscopes with the accelerometer and magnetometer readings to provide stability for the long-term for inclination and heading respectively. The above-mentioned magnetic distortions will affect the heading estimates yielding inconsistent tracking of the motion. Despite the huge improvements in accuracy of orientation tracking over the last years, there are fundamental limits to the accuracy that can be obtained using gyroscopes, accelerometers and magnetometers alone.

In the past years, Xsens spent tremendous effort towards a new motion capture engine that aims to overcome the major sources of error present in current solutions and to provide an accurate and consistent solution. Contrary to the current MVN system, Xsens Analyze/Animate uses a sophisticated motion capture engine that combines the data of all motion trackers with an advanced biomechanical model to get rid of the effects of magnetic distortions. The engine works well in any environment and with a variety of dynamics. This means that measurements can include motions ranging from semi- static to high dynamics.

An important aspect of using any measurement apparatus is the scale of the sensor. Often, for example with a temperature sensor (thermometer!) when the range is exceeded, the sensor either breaks or simply can no longer measure outside of the defined range. With the Xsens sensors and the new engine, this is completely prevented. Clipping (exceeding the dynamic range) is of course possible, but the effect is negligible.

22.6.1.0 Height tracking

Due to the many improvements, from signal processing to how data is handled in Xsens Analyze/Animate including foot contact detection, it is now possible to track height changes of a subject e.g. walking upstairs or up a steep slope while wearing the system. Please note that height changes due to e.g. changing floor level in an elevator will not be detected, since there is no relative change in motion of the body segments, in this case, it will appear that the subject stands still and continues on a level plane.

22.6.2.0 Engine settling time

Settling time is the time needed for the engine to “warm up”. After applying the calibration, or when manually reinitializing the motion capture engine (both when standing still), it is recommended to move slowly (e.g. walk round) for about 30 seconds.

g. Update Rate versus Sample Frequency

Since MVN Studio 4.0, Xsens refers to the term “update rate” rather than sample frequency. This is an intentional change in terminology since the data is sampled in a different way than in the past.

Normal digital systems using some form of recording rely on sampling of this data. Since data is sampled, this implies some degree of data loss at those time instances where no sampling took place. Sampling must take place at a measurable time, some number of times over a given time frame. This is known as the sampling frequency. Most systems operate at a given sample frequency.

The intentional difference in terminology is needed to highlight that data from new Xsens motion trackers (MTx and MTw) undergo a mathematical method called Strap Down Integration. This ensures that no loss of accuracy is experienced, even though the data is transmitted to the PC at lower update rates. Update rates as low as 20 or 30Hz still provide very accurate data.

h. Strap Down Integration

Strap down integration (SDI) is a method used to compute an orientation/position change given an angular velocity and linear acceleration of a rigid body. Angular velocity and acceleration data in the MTx and MTw is sampled at a very high frequency (1000Hz) to maintain accuracy under very dynamic conditions such as vibrations and impacts. The sampling frequency is too high to be transmitted wirelessly and would typically present a computational load that is too high on the receiving host device (e.g. the PC). Based on this data, the MTx and MTw calculate velocity increments and orientation increments using the SDI algorithm. The main advantage of using the SDI algorithm is that full 3D tracking accuracy can be maintained even if the output update rate from the MTx and MTw is lower (<100 Hz). An additional advantage is that the MTx and MTw can continue to track the 3D motion internally when facing transient data loss in the RF transmission and immediately report the full 3D velocity and orientation increment as soon as the RF link is restored, without the need to re-transmit all the data that was lost which would a) cause a large delay during real-time tracking b) use a lot of precious RF bandwidth and consume unnecessary power. The specific use of SDI data in combination with such a specialized RF protocol, Xsens has named the Awinda protocol⁷.

i. The Awinda Protocol

The Awinda protocol is the wireless protocol that controls the data stream from the MTws to the Awinda Station or Awinda Dongle. This wireless protocol provides accurate time synchronization of up to 32 MTw's across the wireless network to within 10 microseconds, achieving 'wired like' system performance. The Awinda protocol is developed such that no data is ever lost. If individual packets are missed, there is some room to contain data temporarily. This buffered data is then retransmitted. If these packets become lost (for example if the subject wearing the hardware is outside of the radio transmission range and for a prolonged period – meaning the buffer becomes full) then it is still possible to retrace the missing data due to the use of SDI. Retransmitted data is used during reprocessing, for this reason Awinda data can appear better – smoother and more accurate after reprocessing.

22.7.1.0 Choosing a Radio Channel for MVN Awinda

Figure 94 provides an overview of the allowed channels for operating MVN Awinda. The Awinda protocol is based on the IEEE 802.15.4, around 2.4 GHz. The bottom row of the figure shows the channels on the 2.4GHz, the top row shows how WiFi channels use the spectrum. This should indicate to the user that the best channels to use when you know which channel WiFi is on. When in an environment where WiFi is also expected to be in prevalent use, but you are not sure which channels, try Channels 11, 15, 20 or 25. Bluetooth uses the entire spectrum around 2.4 GHz but will (try to) avoid channels in use by other systems including Awinda channels.

⁷ Patents pending.

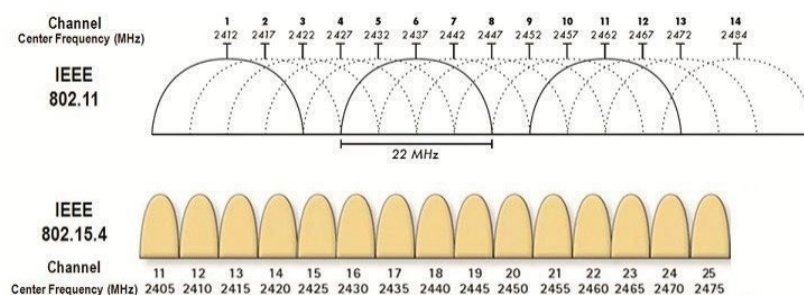


Figure 94: Overview of channels operating around 2.4GHz; for ease of channel selection

j. System Specifications

Recommended computer system

Operating system	Windows 10/11 (64 bit)
Processor	Quad core or higher (2.7 GHz or faster)
Hard drive	SSD 256 GB or more
Memory	8 GB RAM (64-bit)
Graphics card	Hardware acceleration for DirectX 11 Dedicated memory 512 MB or more
Network card	Gigabit Ethernet, support for jumbo frames (9014 Bytes)
USB ports	USB 2.0 (1 per MVN Awinda system, 1 per license dongle)

MVN System

Number of trackers per suit configuration:

Full body	17	inertial motion trackers
Extra prop /backup	1	inertial motion tracker
Optional extra trackers:	3	inertial motion trackers
Total usable prop trackers	4	inertial motion trackers
MVN Camera	Prosilica GS650	
MVN Mounting Straps	Fabrifoam with Velcro closure	
Full body Lycra suit:	Light weight revolutionary stretch fabric with exterior zippered channels for trackers and wires	
Sizes:	Straps one size fits all, with 3 sizes of vests (s-m, m-l, l-xl). Suit: S, M, L (standard), XL, XXL	
Accessories:	2 gloves with tracker pockets 1 head band with tracker pocket 2 foot pads with Velcro patch Pair of shorts (MVN Link)	
MVN Link on body cabling:	1 cable connects Body Pack to Battery Back, all remaining strings of trackers connect to Body Pack.	
Data control units:	1 Body Pack	
Power (BP)		
Battery	Part number: NC2040xx31 (3.1Ah) rechargeable smart Lithium-Ion battery pack	
Battery charger (included)	Part number: CH5000 battery charger with ability to recalibrate fuel gauge on battery packs	
Battery charger power adapter (incl.)	Part number: ATS065TP240 110-240VAC / 24VDC 2.71A (65W)	
Operating time (typical)	9.5 hours	
Power Awinda Station/Charger Power adapter (incl.)	Power adapter Mean Well GE18I12-P1J 90-264VAC /12VDC 1.5A (18W)	

Communication			
Interface	Wireless (WiFi or Awinda Protocol)		
Wireless range radius (max)	MVN	Awinda Station	Awinda dongle
Outdoor	150 meter (492 ft.)	~50m (164 ft.)	~20m
Indoor open space	150 meter (492 ft.)	~20m ()	~10m
Indoor office	50 meter (164 ft.)	~20m ()	~10m

Wireless receiver units:	MVN Link	MVN Awinda
Number of Wireless Receivers	1 Access Point	1 Awinda Station or Awinda Dongle
RF technology	WiFi 2.4GHz, 5.0 GHz	IEEE 802.15.4 PHY 2.4GHz
Interface	Ethernet	USB 1.1 or 2.0
Power	Battery	USB (station additionally mains power supply)

Physical		
Dimensions MTx	36 x 24.5 x 10mm	
Dimensions BP	160 x 72.5 x 25 mm	
Dimensions MTw	47 x 30 x 13 mm	
Dimensions MTw2	49.3 x 30.6 x 13 mm	
Dimensions Awinda Station	148x104x61.9 mm	
Dimensions Awinda Dongle	45x20.4x10 mm	
Dimensions Awinda Dongle ANT	148 x 104 x 62 mm (excluding antenna)	
Dimensions MVN suitcase	559 x 351 x 229 mm (22 x 13 x 9 inches) Flight case with wheels & extendable handle. Durable & watertight	
Weight MTx	10 g	
Weight BP	150 g	
Weight MTw	16g	
Weight MTw	21g	
Weight Awinda Station	233g	
Weight Awinda dongle	8g	
Weight MVN Suit(no trackers)	485 g	
Weight MVN Straps(no trackers)	153 g	
Weight total on-body system (with batteries and cables)	1130 g	MVN Awinda: 525 g
Shipping weight	10 kg	5.5 kg
Housing material products	PC ABS – UL 94V-0	

Environmental

Operation environment	0 – +50 deg Celsius, non-wet
Humidity	20-80%RH NC

Tracker Performance

Static accuracy (Roll/Pitch)	0.2 deg
Static Accuracy (Heading)	0.5 deg
Dynamic Accuracy	1 deg RMS
Accelerometer range	MTx: $\pm 160 \text{ m/s}^2$ (16 g) MTw: $\pm 160 \text{ m/s}^2$ (16g)
Gyroscope range	$\pm 2000 \text{ deg/s}$

MVN Performance

MVN human model	MVN uses a 23 segment biomechanical model with 22 joints. Each joint is specified by statistical parameters for 6DOF joint laxity. An advanced spine and shoulder model is used that computes the kinematics of the spine and shoulder blades.
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System Calibration

General	Flexible calibration scheme with instant feedback regarding the expected accuracy. Calibration can be done without assistance from a second person.
Minimum calibration time	10 seconds. Basic calibration needs only subject length, foot length.
Advanced calibration	10 – 30 seconds per additional step. Advanced subject specific calibration determines tracker alignment and/or subject specific dimensions. Calibration procedures for subjects with limited range of motion are possible.

MVN Fusion Engine

Output	Full kinematics of each segment (position, velocity, acceleration, orientation, angular velocity and angular acceleration). Xsens Animate Pro additionally provides joint angle and CoM. All at preset sample frequency (max 240Hz). Double integration of body segment accelerations allows for jumping/running (permanent floor contact is not needed).
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MVN Fusion Engine

3D translation capture	~1% error in traveled distance (without external aiding). Advanced external contact model detects body-world contacts, to enable crawling, sitting, cartwheel etc. Various friction models (slippage etc.) are possible. ⁸ Seamless fusion with aiding technologies possible, e.g. 3D position from optical/GPS, or 2D image correspondences. ⁸
Magnetic environment Local, permanent disturbances	Full immunity to any magnetic distortion.
Soft tissue artifacts	Minimized to ~2 degrees RMS using redundancy in measurement and biomechanical constraints.
Multiple person capture	Up to four person capturing on one pc

Xsens Analyze/Animate Software

Functionality	Easy and quick calibration Real time preview of motion capture Simultaneous recording and viewing of motion capture data Replaying and editing of previously recorded motion data
Supported export formats	.BVH -Biovision Hierarchical Data .C3D -Coordinate 3D data .FBX – FiLMBOX .MVNX – MVN Open XML format Excel – to open with MS office Excel
Sample Frequency	60 – 240 Hz (default 240 Hz)
Wireless Update Rate	20-60Hz Depending on suit configuration
Data rate	37 MB/min @ 60 Hz 74 MB/min @ 120 Hz 158 MB/min @ 240 Hz

Legal Notice

Warranty	2 Years
MVN and Xsens are trademarks of Xsens Technologies B.V.	
Patents Pending	
www.xsens.com	

⁸ Some features are only available in SDK, ask for possibilities.